

Impinj Speedway
Revolution R220

Impinj Speedway R220 is a high-performance 2-port fixed UHF RFID reader. With patented Autopilot capability for automatic performance optimization, GS1/EPCglobal UHF Gen2 (ISO 18000-6C) compliance, antenna hub support for up to 16 read zones, and IEC IP52 environmental sealing, it delivers enterprise-grade reliability for item visibility solutions.



UHF RFID

Air Interface Protocol	GS1/EPCglobal UHF Gen2 (ISO 18000-6C) or RAIN RFID
Operating Frequencies	FCC: 902-928 MHz; EU1: 865-868 MHz; GX1: 902-928 MHz; GX2: 902-925 MHz
Transmit Power (dBm max conducted)	FCC, GX1, GX2: 32.5 AC / 31.5 PoE; EU1: 31.5 AC / 30.0 PoE
Antenna Ports	2
Maximum Read Rate	200 tags/s
Gen2 Reader Modes	5
Antenna Hub Support	Yes (up to 16 read zones with 2 hubs)
Receive Sensitivity (max)	-84 dBm
Return Loss (min)	10 dB
Antenna Impedance	50 ohm

CONNECTIVITY

Network Connectivity	10/100BASE-T Ethernet
USB Ports	1 device, 1 host
Management Console Port (RS-232)	RJ-45
GPIO Serial Port	Yes
Power Sources	AC-DC power supply; IEEE 802.3af PoE

MECHANICAL

Dimensions (H x W x D)	7.5 x 6.9 x 1.2 in (19 x 17.5 x 3 cm)
Weight	1.5 lb (0.7 kg)
Factory Reset Button	Yes

ENVIRONMENTAL

Operating Temperature	-4 °F to +122 °F (-20 °C to 50 °C)
Storage Temperature	-4 °F to 140 °F (-20 °C to 60 °C)
Humidity	5% to 95% non-condensing
Environmental Sealing	IEC IP52
Shock and Vibration	MIL-STD-810G
RoHS Compliant	Yes
Reliability	Enterprise Grade

SOFTWARE

Network Protocols	SSH, HTTP, HTTPS, NTP, DHCP, SFTP, mDNS
Networking Stack	IPv4, IPv6
Management	SNMPv1, v2, v3
Security	802.1x (Port Security), TLS 1.2 for Secure LLRP
Host Interface	LLRP with Impinj Extensions
Development Libraries	Octane SDK (.NET, Java), LTK (C, C++, .NET, Java), ETK (C, C++)
Software supported	Impinj ItemTest, Impinj ItemSense, Impinj Speedway Connect

FEATURE HIGHLIGHTS

- Patented Autopilot for automatic performance optimization
- High read rates resistant to RF noise and interference
- 2 antenna ports, expandable to 16 read zones with antenna hub
- Power over Ethernet (PoE) support for simplified deployment
- Enterprise-grade reliability and security
- Largest installed base of fixed readers across the industry

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed UHF RFID Reader